LATEST 15-BUTTON KEYMAP

Default layer and physical button numbering

NUMBERED PHYSICAL BUTTONS

1. **Ctrl+C** — Tap; hold for Fine Cursor Movement
2. **Ctrl+V** — Tap; hold for Drag Scroll
3. **Delete** — Keyboard Delete
4. **Enter** — Keyboard Enter
5. **Left click** — Mouse button 1
6. **Right click** — Mouse button 2
7. **Middle click** — Mouse button 3
8. **Screenshot** — Win+Shift+S
9. **Control Settings** — Hold to access layer
10. **Drag Lock** — Tap to hold click; tap again to release
11. **Unassigned** — Available in ZMK Studio
12. **Key diagnostic** — Types /;.l,kmj?:>L<KMJ then Enter
13. **Feedback Controls** — Hold to access layer
14. **Status Report** — Hold to access layer
15. **Device Settings** — Hold to access layer

LAYER MAP

The exact ZMK Studio layer order and purpose

0 **Default**
Everyday clicks, shortcuts, layer holds and Drag Lock

1 **Control Settings**
Pointer and twist sensitivity; Bluetooth profile selection

2 **Device Settings**
Power, Studio, scroll mode, Bluetooth clearing and reset

3 **Status Report**
Reports sensitivity, endpoint, scroll mode and battery

4 **Feedback Controls**
Persistent vibration and LED enable/disable toggles

5 **Drag Scroll**
Momentary ball-to-scroll conversion while button 2 is held

6 **Scroll Mode Switch**
Internal layer active whenever standard full-notch mode is selected

7 **Fine Cursor Movement**
Momentary fine pointer movement while button 1 is held

Priority note

Layers 5-7 primarily change trackball processing. Their transparent key bindings allow normal buttons and held configuration layers to continue working.

CONTROL SETTINGS

Hold button 9 (Left A), then press an action button

HOLD 9 + ACTION

9 **Layer hold**
Keep held while choosing an action

10 **Twist sensitivity +**
Increase twist-scroll sensitivity

11 **Twist sensitivity -**
Decrease twist-scroll sensitivity

12 **Bluetooth next**
Select next Bluetooth profile

13 **Bluetooth previous**
Select previous Bluetooth profile

14 **Pointer sensitivity +**
Increase pointer sensitivity

15 **Pointer sensitivity -**
Decrease pointer sensitivity

Sensitivity range

Pointer and twist each have 20 levels. Changes persist after restart. Repeated presses at the minimum or maximum use the boundary warning feedback.

DEVICE SETTINGS

Hold button 15 (IO3), then choose an action

HOLD 15 + ACTION

9 **Power off**
Hold action key 2 seconds

10 **Unlock ZMK Studio**
Tap to enable Studio access

11 **Toggle scroll mode**
Tap: Standard / High-resolution

12 **Clear all Bluetooth**
Hold action key 2 seconds

13 **Clear current Bluetooth**
Hold action key 2 seconds

14 **Reset sensitivities**
Hold 2 sec: pointer 7, twist 5

15 **Layer hold**
Keep held while choosing an action

Important

Keep button 15 held throughout every action. Power-off, Bluetooth clearing and sensitivity reset require a separate two-second hold to prevent accidental activation.

STATUS REPORT

Hold button 14 (IO2), then press a report key

HOLD 14 + REPORT

9 — Twist sensitivity
Blue flash; pulses encode level 1-20

10 — Current Bluetooth profile
Purple flash; profile count or ESB long pulse

11 — Pointer sensitivity
Green flash; pulses encode level 1-20

12 — Scroll mode
Red + long = Standard; green + 3 short = High-res

13 — Battery charge
Green/amber/red; long per 25%, short per 5%

14 — Layer hold
Keep held while choosing a report

How numeric reports work

Sensitivity: each long pulse represents 10 levels; each short pulse represents 1. Battery: each long pulse represents 25%; each short pulse represents the next completed 5%. Example: 40% = 1 long + 3 short.

FEEDBACK CONTROLS

Hold button 13 (IO1), then press a toggle key

HOLD 13 + TOGGLE

9

Toggle vibration

Persistent. Enabling: green flash + 180 ms vibration; disabling: orange flash, then motor off.

10

Toggle LED feedback

Persistent. Enabling: cyan flash; disabling: orange flash, then LEDs off.

13

Layer hold

Keep held while choosing a feedback toggle.

Persistence

Both choices survive power cycles. Turning vibration off also suppresses direct status pulses. Turning LED feedback off blocks RGB output after the confirmation flash.

Notification replacement

A new custom action cancels any unfinished direct Status Report vibration before its own feedback starts, preventing overlapping motor sequences.

MODES & BUTTON IDENTITY

Momentary modes, persistent scroll selection and GPIO mapping

Fine Cursor Movement - hold 1

Pointer movement becomes finer while held. The LED shows a dim, smoothly breathing rainbow.

Drag Scroll - hold 2

Ball movement becomes scrolling while held. The LED shows a dim, smoothly breathing rainbow.

Drag Lock - tap 10

First tap holds left click; second tap releases it. Magenta confirms enabled; cyan confirms disabled.

Scroll mode - hold 15 + tap 11

Standard produces full wheel notches. High-resolution provides smoother twist scrolling. The saved selection survives restart.

BOTTOM ROW - LEFT TO RIGHT

9	10	11	12	13	14	15
Left A	Left B	Right A	Right B	IO1	IO2	IO3
P0.17	P0.16	P1.08	P1.09	P0.11	P0.15	P0.20

Electrical connection

Every added switch is normally open and connects only its assigned signal pin to confirmed GND when pressed. Internal pull-ups are enabled.

Compiled and saved scroll state

Standard full-notch scrolling is the compiled fallback. Once a scroll mode is selected, the saved selection takes priority after restart.

CUSTOM LED & VIBRATION

User-invoked layers, modes, controls and device actions

EVENT	LED FEEDBACK	VIBRATION PATTERN (ms)
Layer: Control Settings	Dim cyan flash	80
Layer: Device Settings	Dim amber flash	80 70 80
Layer: Status Report	Dim blue-cyan flash	Report action supplies vibration
Layer: Feedback Controls	Dim purple flash	-
Fine Cursor Movement held	Dim rainbow, smooth breathing	-
Drag Scroll held	Dim rainbow, smooth breathing	-
Drag Lock enabled	Magenta flash	70 70 220
Drag Lock disabled	Cyan flash	220 70 70
Sensitivity reset	Cyan flash	70 70 70 70 70 180 300
Pointer increased	Green flash	70 70 220
Pointer decreased	Orange flash	220 70 70
Pointer at lowest	Rainbow warning	220 70 70 70 220
Pointer at highest	Magenta blink	350 80 350
Twist increased	Blue flash	70 70 70 70 220
Twist decreased	Yellow flash	220 70 70 70 70
Twist at lowest	Rainbow warning	220 70 70 70 70 70 220
Twist at highest	Cyan blink	350 80 350
Standard selected	Three red flashes	220 70 220 70 70
High-resolution selected	Three green flashes	70 70 220 70 220
ZMK Studio unlocked	Green flash	60 60 60 60 60
Clear current Bluetooth	Orange blink	300 120 70
Clear all Bluetooth	Red blink	300 100 300 100 300
Power off	Red fade	500
Vibration enabled	Green flash	180
Vibration disabled	Orange flash	Motor disabled
LED enabled	Cyan flash	-
LED disabled	Orange flash	-

STATUS REPORT REFERENCE

Hold button 14; report keys are 9 through 13

STATUS REPORT	LED FEEDBACK	VIBRATION ENCODING
Twist sensitivity	Blue flash	Long = 10 levels; short = 1 level
Pointer sensitivity	Green flash	Long = 10 levels; short = 1 level
Bluetooth profiles 1-5	Purple flash	1-5 short pulses (80 ms ON / 85 ms OFF)
ESB dongle	Purple flash	One 600 ms long pulse
Standard scrolling	Red flash	One 300 ms long pulse
High-resolution scrolling	Green flash	Three 90 ms pulses
Battery above 50%	Green flash	Long per 25%; short per remaining 5%
Battery 21-50%	Amber flash	Long per 25%; short per remaining 5%
Battery 0-20%	Red flash	Long per 25%; short per remaining 5%
Battery example: 0-4%	Red flash	No pulse: no completed 5% step
Battery example: 5%	Red flash	1 short
Battery example: 20%	Red flash	4 short
Battery example: 25%	Amber flash	1 long
Battery example: 40%	Amber flash	1 long + 3 short
Battery example: 50%	Amber flash	2 long
Battery example: 75%	Green flash	3 long
Battery example: 100%	Green flash	4 long

SYSTEM LED & VIBRATION

Automatic connection, battery, profile and hardware indications

SYSTEM EVENT	LED FEEDBACK	VIBRATION PATTERN (ms)
USB connected	Green flash	40 50 100
USB disconnected	Orange flash	100 50 40
Advertising MUI	Amber flash	35 100 35
Bluetooth profile 1	Profile color	220 180 60
Bluetooth profile 2	Profile color	220 180 60 60 60
Bluetooth profile 3	Profile color	220 180 60 60 60 60 60
Bluetooth profile 4	Profile color	220 180 60 60 60 60 60 60 60
Bluetooth profile 5	Profile color	60 60 60 60 60 60 60 60 60
ESB dongle selected	Temporary rainbow	220 100 220
No endpoint available	Red flash	500 150 500
Battery warning 1	Yellow flash	100 100 100
Battery warning 2	Orange flash	100 80 100 80 100
Battery warning 3	Red-orange flash	180 100 180 100 180
Battery critical 1	Red flash	300 120 300
Battery critical 2	Red flash	300 100 300 100 300
Battery critical 3	Red flash	400 100 400 100 400
OS type toggled	Cyan flash	40 40 40 40 120
Keymap slot switched	Magenta flash	120 40 40 40 40
Key auto-held	No dedicated light	35
Left sensor error	Continuous red blink	350 100 350 180 70
Right sensor error	Continuous red blink	350 100 350 180 70 70 70